

Relational Algebra $\Rightarrow$

- $\rightarrow$ Relational algebra is a procedural query language which takes the instance of a relation as input and yields instance of relation as output. It gives the step by step to obtain the result of the query.
- $\rightarrow$ Relational algebra mainly provides theoretical foundation for relational databases and SQL. It uses operators to perform queries.

Selection Operation (σ) $\Rightarrow$

- $\rightarrow$ The selection operation is used to select a subset of the tuples from a relation that satisfies a select condⁿ. σ is used to represent selection operation.
- $\rightarrow$ The select operator is unary such that it is applied to a single relation. The general format for select operation is:

$\sigma_{\text{condition}}(R)$

where, σ is used for selection prediction.

R is used for relation.

select condition is the relational operator.

e.g. write a query to give all the information of student having Roll no. = 4.

$\Rightarrow$ Query: $\sigma_{\text{Roll no.}=4}(\text{Student})$.

Q. Find all the information of student having Name = 'Arun' and Address = 'Delhi'.

$\Rightarrow$ Query: $\sigma_{\text{Name}='Arun' \ \& \ \text{Address}='Delhi'}(\text{Student})$.

or $\sigma_{(\text{Name}='Arun') \vee (\text{Address}='Delhi')}(\text{Student})$

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Project Operation $\Rightarrow (\pi)$

- $\rightarrow$ Project operation selects a certain columns from the table and discards other columns. This operation shows the list of those attributes that we wish to appear in the result. Rest of the attributes are eliminated from the table.
- $\rightarrow$ The general format:

$\pi_{\text{Attributes}}(R)$

where, π represents the project operation and

Attributes is the list of attributes from the attributes of relation R .

e.g. $\Rightarrow$ Find the student name from the student relation.

$\Rightarrow$ Query: $\pi_{\text{name}}(\text{Student})$

Q. Find the name and address of the student.

$\Rightarrow$ Query: $\pi_{\text{name, address}}(\text{Student})$.

Union Operation $\Rightarrow$

- $\rightarrow$ Union perform the binary union between two given relations and is defined as: $R \cup S$, where R and S are either database relation or relation result set.

$\rightarrow$ Union operation to be valid, the following condⁿ must hold:

- R and S must have same number of attributes.
- Attribute domain must be compatible, it avoids duplicate values of tuples automatically.

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E.g. $\rightarrow$ write a relational algebra query for combining the name of student from the student1 and student2 relation.

$\Rightarrow$ Query: $\pi_{\text{name}}(\text{Student1}) \cup \pi_{\text{name}}(\text{Student2})$.

Intersection Operator $\cap$

$\Rightarrow$ It performs the binary intersection between two given relations and is defined as $R \cap S$.

E.g. $\rightarrow$ write query for intersecting for the name of student from the student1 and student2 relation.

$\Rightarrow$ Query: $\pi_{\text{name}}(\text{Student1}) \cap \pi_{\text{name}}(\text{Student2})$.

Set difference operation $-$

$\Rightarrow$ The result of set difference operation is tuples which are present in one relation but are not in second relation.
Syntax: $R - S$

E.g. $\rightarrow$ Find all the tuples which is the name of the author who have written books but not articles.

$\Rightarrow$ $\pi_{\text{author}}(\text{Books}) - \pi_{\text{author}}(\text{Articles})$.

Cartesian Product $\times$

$\Rightarrow$ It combines the information of two different relations into one. Notation: $R \times S$

where, R and S are relations:

$\Rightarrow$ And their output will be defined as

$$R \times S = \{ \langle t_1, t_2 \rangle \mid t_1 \in R \text{ and } t_2 \in S \}$$

Rename Operation ρ

$\Rightarrow$ The result of relational algebra are also relations but without any name, the rename operation allows us to rename the output relation.

$\Rightarrow$ Rename operation is denoted with ρ .

Notation: $\rho_x E$

where, the resultant expression E is named with name x .